

Curvature Variable Physics (CVP): Eliminating Divergence in Artemis Lunar Descent Through Dynamic Curvature Modeling

Timothy J. Dillon
206 Innovation Inc., Bellevue, WA

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Abstract

This manuscript presents a formal mathematical demonstration that static-curvature gravitational modeling is insufficient for stable Artemis terminal descent, and that a curvature-variable correction eliminates the divergence term that destabilizes the final descent corridor. The framework preserves the original lunar-descent focus while adding a tightly scoped cross-domain outlook on magnetospheric stability and a concise computational implementation note. Dynamic curvature response, governed by the Dillon Equation, restores stability under mascon-rich conditions.

Keywords. Curvature Variable Physics; Artemis lunar descent; mascon modeling; Dillon Equation; divergence cancellation; Omega-Genesis Framework.

1 Introduction

Crewed lunar landing requires gravitational modeling fidelity beyond smooth two-body assumptions. In the final descent corridor, mascons, local heterogeneity, and anisotropic curvature gradients dominate stability. CVP treats curvature as a dynamic field and restores the terms omitted by the static baseline.

Curvature-sensitive propagation is expressed via the Dillon Equation:

$$c_{\text{eff}} = f(K, R, \nabla K, \partial_t K),$$

with response operator $C(K) = \beta_1 \partial K / \partial t + \beta_2 \nabla K + \beta_3 \nabla^2 K$.

2 Static-Curvature Failure Mode

Under the classical potential $\Phi(r) = -GM/r$ and associated curvature $K = 2GM/r^3$, the baseline assumption $\nabla K = 0$ removes the very gradient terms that become decisive near the surface. The omitted force produces a divergence operator that grows rapidly as the vehicle approaches the surface, preventing a stable curvature-consistent landing solution.

3 CVP Correction

CVP promotes curvature to $K = K(t, r, \theta, \phi)$ and introduces the response operator $C(K)$. Coefficients are selected to cancel the destabilizing divergence term, yielding $D_{\text{CVP}} = 0$ as the convergence condition. The Syntropy Operator $\Phi = -C(K)$ enforces the compatibility condition $\Delta + \Phi = 0$.

4 Simulation Summary

Across tested conditions, the static-curvature model fails to converge, while the CVP model remains bounded and convergent. Representative results include terminal velocity residuals below 0.3 m/s and radial error below 0.1 m under the CVP-corrected formulation. Detailed robustness analysis under controlled dissipative perturbations is recorded in the appendix.

5 Mascon-Aware Extension

Mascons are the dominant source of lunar curvature gradients. In the CVP framing, mascon perturbations are treated as first-class features of the dynamic field. High-fidelity GRAIL products (GRGM1200A, GRGM1500E) inform the model.

Mascon perturbations are modeled as depth-weighted additions:

$$K_{\text{mascon}}(r) = 0.35 \exp\left(-\frac{(r - r_{\text{surface}} - 40000)^2}{10^{10}}\right) - 0.18 \exp\left(-\frac{(r - r_{\text{surface}} - 10000)^2}{2 \times 10^{10}}\right).$$

The response operator automatically absorbs these structures, converting destabilizing anomalies into stabilizing corrections.

6 Computational Implementation Note

The analytical framework is compatible with a tensor-based GPU implementation in which a high-resolution curvature field is sampled, ∇K and $\nabla^2 K$ are evaluated numerically via tensor gradients, the CVP response operator is assembled in real time, and the descent state is advanced under a convergent update rule. This enables edge-AI implementation paths for Artemis-class landers.

7 Transition from Orbital Approach to Terminal Descent

The manuscript focuses on the terminal descent corridor. At higher altitudes, conventional models remain useful. The handover marks the regime where curvature must be treated dynamically. CVP serves as a refinement layer above the handover and becomes part of the governing descent law below it.

A Computational Verification and Reproducibility

Detailed simulation results, phase-plane behavior, and robustness analysis under mascon perturbations are available in the companion binder.

B References

C Dependency Discipline and Non-Circular Structure

The logical dependency order is one-way: static failure analysis $\rightarrow$ CVP correction $\rightarrow$ mascon-aware extension $\rightarrow$ implementation closure.